

VegaHealth

February 23, 2026

The Honorable Thomas Keane, M.D.
Assistant Secretary for Technology Policy
U.S. Department of Health and Human Services
220 C St SW
Washington, DC 20201

Re: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Dr. Keane,

Vega Health appreciates the opportunity to provide comments to the Assistant Secretary for Technology Policy's/Office of the National Coordinator for Health IT's (ASTP/ONC's) request for information on accelerating the adoption and use of clinical AI. Vega Health supports the Administration's efforts to make safe, secure, and effective clinical AI more readily available to the healthcare industry. In our detailed comments below, Vega Health will focus on the last mile between regulatory intent and real-world integration of new technology into the healthcare ecosystem. Vega Health's business model is dedicated to ensuring widespread health system access to the most effective, validated AI solutions. We are deeply aligned with ASTP/ONC's mission in this space, and our leadership understands the barriers that exist to scale effective healthcare AI. As the agency seeks to accelerate adoption and use of this technology, we specifically urge ASTP/ONC to consider the following:

- The role of federal deregulation and harmonization efforts in accelerating adoption of innovative clinical AI solutions;
- Pathways to accelerating clinical AI adoption through partnerships with state sandbox initiatives and market-driven incentives;
- Approaches to overcoming the obstacles limiting clinical AI implementation at every layer of the healthcare AI stack;
- Lessons learned from decades of clinical AI implementation and evaluation at leading health systems around the U.S.;

- Strategies to promote interoperability and eliminate information blocking practices that inhibit innovation; and
- Approaches to clinical AI research that can meaningfully accelerate adoption for health systems.

Vega Health appreciates ASTP/ONC's commitment to accelerating the adoption of clinical AI to drive better patient outcomes and improve the sustainability of American healthcare.

I. Background on Vega Health

Vega Health is dedicated to helping healthcare organizations realize measurable value from healthcare AI. We operate a proprietary platform that connects health systems with a curated marketplace of validated AI solutions. We also provide comprehensive integration support and objective performance monitoring of AI solutions. Our mission is to be the objective, trusted partner equipping health systems everywhere as they scale AI solutions that improve care and operational outcomes.

For health systems, Vega Health provides:

- A proprietary platform that equips healthcare organizations to efficiently run multiple AI solutions within their digital environment;
- A curated marketplace of AI solutions validated in real-world healthcare settings;
- Expert development and implementation support;
- Rigorous measurement and monitoring of clinical outcomes and operational impact; and
- Partnership in co-developing AI solutions optimized for local patient populations.

For healthcare innovators, Vega Health offers:

- Commercialization pathways to scale AI solutions beyond sites of origin;
- Distribution channels to reach health systems seeking validated solutions;
- Implementation expertise to ensure successful deployments in new environments; and
- Transparent monitoring evaluating effectiveness in real-world settings.

Vega Health's approach to integrating and monitoring AI solutions is built on more than a decade of real-world experience at the Duke Institute for Health Innovation (DIHI) and through the Health AI Partnership (HAIP). At DIHI, our CEO and co-founder Dr. Mark Sendak worked to develop, integrate, and monitor more than 60 AI solutions in live clinical settings at Duke Health. This work included the development of Sepsis Watch, still considered one of the most successful real-world healthcare AI integrations as documented in the peer-reviewed literature.

The work of DIHI and HAIP established foundational frameworks that now guide responsible AI adoption and monitoring across the healthcare industry, including widely cited guidelines for clinical AI integration (the HEAAL framework). HAIP's work to share best practice frameworks and build an effective practice network beyond academic medical centers – including community hospitals, Federally Qualified Health Centers (FQHCs), and rural health systems – highlighted many of the practical challenges of AI implementation across diverse environments. Vega Health exists to bridge these critical gaps. Many healthcare organizations lack the infrastructure, expertise, and resources to source, integrate, and evaluate the effectiveness of AI solutions. This creates a systemic risk as AI adoption accelerates, threatening to widen a 'digital divide' where one class of health systems leverages AI while a second group falls farther behind. We applaud the ASTP/ONC's work to reduce regulatory burden for innovators seeking to assist these healthcare organizations and look forward to this and future opportunities to engage with the agency and its leadership.

Vega Health's business model operationalizes many of the frameworks promulgated by DIHI and HAIP. Through our platform architecture and monitoring approach, we guide health systems through objective evaluation of technical AI model performance, user adoption and behavior change, and outcome measures. We are committed to acting as an objective third party that provides unbiased performance data to both health systems utilizing AI solutions, as well as the developers of those solutions, all with the goal of advancing innovation.

Vega Health measures performance across four dimensions: technical accuracy/fidelity, user adoption patterns, achievement of clinical/operational outcomes, and return on investment. We implement comprehensive monitoring strategies which use real-world evidence to determine whether an AI solution is delivering value in practice. This approach ensures that performance is evaluated not only in aggregate, but across patient subpopulations. Our team has extensive experience working in clinical settings to evaluate and monitor real-world AI performance, rather than operating at the level of theoretical frameworks.

II. Barriers to Innovation (Question 1)

Vega Health strongly supports efforts across the Department of Health and Human Services (HHS) to simplify the duplicative and burdensome regulatory thicket that is currently holding back innovative healthcare solutions from scaling and delivering value. Currently, the state [AI law landscape](#) is disjointed and daunting: As of this writing, 46 states have enacted 284 bills, and 2026 has already seen 771 bills introduced across 31 states. Despite [federal efforts](#) to limit the scope of state AI legislation, the current landscape forces innovators to spend time navigating complex regulations.

There are two well-established pathways for AI regulation: development-based regulation and use-based regulation. As [Andreessen Horowitz](#), the [Consumer Technology Association \(CTA\)](#), and others have argued, layering regulations on the manner in which AI models can be developed will only serve to increase compliance costs, make it more expensive to develop and scale AI solutions in multiple states, and create a digital ecosystem where only entrenched market incumbents have the scale, budget, and legal team necessary to successfully scale AI solutions. By eliminating regulatory burden and focusing enforcement on already-unlawful outcomes, particularly in heavily regulated sectors like healthcare, the Administration will further incentivize innovative solutions to broken processes that can reduce costs and improve health outcomes for patients and providers.

As Andreessen notes in the organization's [Roadmap for Federal AI Legislation](#), state and federal regulation must focus on harmful uses of AI, not preemptively impose barriers that limit market access. Developer-facing regulations drastically extend the technology development phase and extend the time to market entry, making it significantly more difficult for fast-moving innovators to bring solutions to market and challenge incumbents. A competitive health AI market drives overall improvements to privacy, cybersecurity, and safety by forcing companies to compete to deliver the best product – and does not preclude current or future regulations from heavily penalizing realized harms, failures, or misuse.

This regulatory simplification is particularly important for new market entrants and innovators. Vega Health's business is built on the ability to scale the most effective AI models – technologies developed by leading Academic Medical Centers and already validated in clinical settings around the country – to solve complex problems for health systems of all sizes. The Vega Health marketplace hosts models developed by clinical experts that improve clinical and operational workflows for patients and providers. The Vega Health marketplace already includes models to predict patient deterioration, post-surgery complications, emergency department triage, and in-hospital mortality.

The recent HTI-5 Notice of Proposed Rulemaking (NPRM) takes important steps towards simplifying and aligning the security, interoperability, and privacy requirements broadly underpinning health IT. ASTP/ONC is helping to lay the groundwork for widespread AI innovation, development, and scaled adoption in healthcare settings of all sizes – including the agencies' collective focus on ending anti-competitive practices by industry incumbents.

Vega Health applauds ASTP/ONC's deliberate efforts to reduce the regulatory, administrative, and invented hurdles that have been leveraged by large incumbents to disadvantage new market entrants, particularly those contracted by health systems to improve outcomes by leveraging data contained within the electronic health record. Complex and conflicting regulations governing AI, privacy, and interoperability have been leveraged by some certified health IT vendors to obstruct this data access or impose undue financial burden to utilize this data.

III. Incentives (Question 2)

The healthcare sector has one of the [highest adoption rates](#) of commercial AI, led by a [71% adoption rate](#) among health systems. At the physician level, the [American Medical Association reports](#) broad belief that AI will improve efficiency and burnout, diagnostics, clinical outcomes, and even patient safety.

However, the uptake of clinical AI – and its potential benefits – is not distributed equally among health systems. While 53% of health systems report high success with AI scribes, adoption remains [sharply divided](#): 86% among system-affiliated hospitals versus just 37% among independent facilities. Rural hospitals [lag even further](#), with only 56% reporting any AI integration compared to 81% of urban hospitals.

Across the United States, regulatory sandboxes have emerged as a key strategy to accelerate AI innovation under necessary guardrails and oversight mechanisms. To date, [Utah](#), [Texas](#), and [Delaware](#) have developed AI sandboxes that include healthcare or healthcare-adjacent sectors; notably, Utah also recently [announced](#) a major pilot program with a health AI vendor to use AI to refill prescription medications.

Despite this transformative potential, major health systems have yet to widely participate in AI sandbox initiatives, slowing adoption of clinical AI solutions. This is largely due to the nature of healthcare AI regulation – providers subject to state and federal laws need certainty on both to comfortably pilot innovative clinical solutions. Apart from disclosure requirements and medical licensing protections, [state AI](#) and [data privacy laws](#) primarily govern direct-to-consumer healthcare occurring outside of health systems – leaving a large grey area for clinical AI solutions that may be subject to federal regulations and guidance.

There is a degree of urgency to bringing providers into health AI pilots and sandbox initiatives: Without direct participation from provider organizations, AI vendors may [divert patients](#) from physicians and undermine the local practice of medicine.

To increase healthcare provider participation in AI sandbox initiatives, there must be incentives that reduce the regulatory risk inhibiting internal health system innovation. The regulatory requirement imposing the most risk and burden on these health systems is the Food and Drug Administration's (FDA's) [Guidance Document](#) on Clinical Decision Support (CDS) Software, which establishes criteria under which CDS software is regulated as a medical device. While recent updates expanded federal enforcement discretion, the risk of device classification still dampens investment in health system AI innovations. The FDA can partner with states in the design of certain healthcare-focused sandbox agreements to provide health systems with more robust regulatory relief; in exchange, the states can provide the FDA with real-world AI evaluation and monitoring evidence for novel CDS applications as part of the contract framework for AI sandbox participation.

The proposed framework would incentivize health system participation in AI sandbox initiatives while advancing evidence-based AI policy nationally. By requiring safety and efficacy evidence to be generated as part of sandbox participation, the states could offer the FDA a way to close a [crucial gap](#) in post-implementation monitoring of clinical AI. Finally, the guardrails around sandbox participation could significantly advance safety measures, cybersecurity, and reporting requirements currently hindering adoption of clinical AI. This partnership offers a pathway to accelerated AI innovation and adoption in clinical settings, transforming care for a high volume of patients in traditional healthcare settings.

Furthermore, Vega Health urges ASTP/ONC to consider how market-based incentives shape AI innovation and evaluation – a stark contrast to top-down “model card” and pre-deployment requirements.

Industry efforts to compile AI “model cards” containing common data elements have not worked as intended. Vega Health's Co-Founder and CEO published the first Model Facts label for machine learning models used by healthcare systems. Since that time, DIHI and HAIP have facilitated the adoption of Model Facts labels by health systems to improve communication with clinical front-line users. In contrast to this bottom-up approach, other efforts to centralize “model cards” have not effectively served end users. Rather, many such efforts became a gatekeeping exercise, with well-resourced industry groups founded by large model developers competing to set the terms of market entry and protect their own incumbent technologies.

Federal standards for the use of AI can protect consumers, but it is a futile exercise to predict how AI model performance on an ambiguous and evolving set of metrics will translate to real-world performance across vastly different clinical settings. Meaningful

regulatory action can best protect patients and consumers by requiring disclosure of safety incidents and punishing actual harm. Vega Health strongly supports ASTP/ONC's effort to eliminate this restriction on competition by removing model cards from the health IT certification criteria.

In the healthcare sector, health systems procure AI technologies to fulfill specific business objectives across administrative, clinical, and quality workflows. In order to remain in compliance with federal regulations, assess organizational success, and ensure the delivery of quality patient care, health systems have the right to require some degree of model transparency and performance evaluation. However, these metrics, the manner of assessment, and the confidentiality of reporting should be contractually defined on a case-by-case basis, not set by conflicted coalitions or used to limit access to app markets.

This does emphasize the need for transparent, objective post-deployment model evaluation and monitoring in healthcare becomes even more important. Rather than adding a federal stamp of approval to the AI model evaluation metrics proposed by special interest groups, ASTP/ONC should encourage health systems and end users to determine the most impactful metrics. Well-established contracting principles will then shape the market and hold developers accountable for delivering on promised performance, just as it has for Software-as-a-Service (SaaS), medical equipment, and countless other markets.

Additionally, model evaluation should advance model procurement, implementation, deployment, and maintenance or sunset decisions by the end user, not present another barrier to market entry. As ASTP/ONC has recognized, federal standards for evaluating and reporting on AI performance are inflexible and unnecessarily burdensome. Not only does the performance of healthcare AI models vary across patient populations and payer mixes, but benchmark and threshold performance will vary by health system. It makes far more sense for health systems to determine what the minimum acceptable model performance should be for their system and evaluate on an ongoing basis than to allow large incumbents to set model evolution criteria to limit market competition.

As the healthcare technology market moves away from AI gatekeepers and towards real-world model evaluation and monitoring, Vega Health is confident that a market-based framework driven by innovation will be able to replace the restrictive model card requirements that ASTP/ONC proposes to remove.

IV. Implementation Challenges & Generating ROI (Questions 3 & 6)

Vega Health has a unique perspective and expertise developed during more than a decade of working directly with health systems to build, integrate, and scale effective AI solutions. First at Duke, and now at multiple regional and multistate health systems, Vega Health has seen a series of recurring implementation challenges at each level of the emerging health AI stack.

Fundamentally, Vega Health categorizes the health AI stack into a series of layers: the database layer, the connection layer, the data mapping layer, the solution layer, and the observability & governance layer. At each stage of AI model or solution development and deployment, several obstacles have prevented pilots from scaling and generating true ROI for health systems.

At the database layer, interoperability challenges persist across real-time FHIR-enabled data, data elements that still require HL7-based exchange (like Admissions, Discharge and Transfer notifications), financial and revenue cycle data, and delayed data that is stored in data lakes or tables. For example, Medication Administration Records (MAR) are critical to acute clinical AI solutions, but this data element is not yet available via FHIR. Vega Health supports ASPT/ONC's proposal to add MAR data to USCDI v7. The range of data sources and formats – as well as local differences in the database architecture, EHR selection, and data-level vendors of each health system – make it immensely challenging to translate AI models from one health system to another. Models often rely on a set of input variables, which must be mapped to a unique set of data elements and fields for each health system. Vega Health is encouraged by the recent HTI-5 NPRM targeting some of the artificial barriers erected by EHR vendors to access this data in service of health system innovation.

This data grouping process is time-intensive and prone to quality assurance failures. For example, each time a health system records a lab or measurement in a different unit or under a different name in the EHR, there must be a process in place to map that new data element to the corresponding standardized input variable for an AI model – a process that repeats for each model running in a health system, each with a potentially unique set of variable names. It is easy for this process to slip, risking model drift and flawed outputs. If a health system does not have a robust real-time evaluation and monitoring mechanism in place, these failures can take weeks or even months to catch – risking incorrect alerts and best and patient safety incidents at worst.

Evaluating and monitoring AI solutions pose additional challenges for even the most technologically mature health systems. Each AI pilot or vendor reports on safety and performance metrics in a different format. Accessing and reviewing myriad individual dashboards and downloadable performance reports complicates real-time model

performance and observability. Even with processes in place to review each model's reported metrics at a regular cadence, it can be nearly impossible for clinical or technical leaders at health systems to catch and correct drift or deteriorating performance in real time. Further complicating the challenge is a disconnect between model performance reporting provided by vendors and the clinical outcomes and workflows impacted by the model outputs. Without a standardized command and control center, each performance report must be translated to determine whether model performance issues affect time, accuracy, frequency, confidence, precision, or several of these elements in the clinical setting. Healthcare AI needs a unified observability layer that can scale with new solutions, whether provided by a third party or developed internally by the health system.

Finally, as health systems across the United States pilot and deploy an ever-increasing number of AI solutions from a range of vendors, cybersecurity challenges continue to proliferate. Health systems need a simple way to pause models for evaluation, take them offline temporarily to resolve security incidents, or even sunset models that become vulnerable legacy systems or fail to perform at an acceptable level. Leaving AI solutions connected to the most crucial data and clinical applications in a health system is an unmitigated vulnerability; as early AI pilots wind down and health systems make long-term vendor selections, securely sunsetting AI solutions will become a key security process in the coming years.

HHS must help health systems ensure that rapid innovation in clinical AI does not exacerbate cybersecurity challenges facing health systems. Valuable patient data, complex digital supply chains and [internet-connected devices](#), costly and legacy technology, and status as critical infrastructure make hospitals and health systems a top target for cyber criminals and foreign threat actors. One 2025 report found that [93% of health systems](#) experienced a cyber incident over a 12-month period, and healthcare data breaches are among the [costliest incidents](#) each year. The scale of this challenge is immense, and the rapid development and implementation of internet-connected AI technology in critical clinical settings does carry some risk. In addition to the technical solution described above, HHS can provide technical assistance to help health systems mitigate cyber risk from clinical AI, [incorporate cybersecurity into emergency preparedness funding programs](#) like the Hospital Preparedness Program, and update cybersecurity best practices – like the [405\(d\) Program](#) – to include guidelines for AI cybersecurity.

V. Evaluating Clinical AI (Questions 4 & 5)

Evaluating clinical AI is a unique challenge at the heart of more widespread adoption. The earliest wave of health AI adoption was across administrative functions, with

evaluation metrics similar to standard SaaS platforms that have been implemented and evaluated in healthcare back offices for the better part of a decade. In the clinical setting, healthcare's digital maturity has long lagged other sectors like finance, manufacturing, and technology. Now, as the adoption of clinical AI [rapidly accelerates](#), healthcare is leapfrogging other sectors that have traditionally led the maturity curve in both AI spending and adoption rates.

However, healthcare's longtime laggard status can provide immense benefit in the evaluation of clinical AI. The tenets of technical AI/ML model performance and optimization have already been well-established in both literature and practice. These technical principles are largely unchanged in the clinical setting. One key lesson learned from the use of AI/ML in advanced manufacturing is that feedback loops must link telemetry to outcomes of interest. In a clinical setting, this means that adjustments to model performance telemetry – accuracy, precision, sensitivity – must translate to measurable changes in target clinical metrics, such as patient outcomes, quality measures, response rates, or time to action.

In order to ensure that model telemetry is closely and accurately linked to clinical outcomes, health AI should follow an implementation, testing, and deployment process grounded in safety and rigorous evaluation. Vega Health's model is built on licensing clinically validated health AI models from leading health systems and deploying them in health systems around the country. However, models that have been evaluated on one data set and one health system's unique patient mix are not guaranteed to operate identically in another health system's environment, or on a demographically different patient population. Therefore, Vega Health has built an implementation process around retrospective evaluation to validate performance, a silent pilot on real-time data from the client patient population to tune telemetry, and ongoing real-time evaluation and monitoring after go-live.

Additionally, clinical model performance must be measurable down to the individual physician or end user level. Technical performance metrics in a vacuum – or even in aggregate system-wide – do not fully convey whether a model is being appropriately integrated into workflows and utilized by end users. Clinicians must understand how to interpret and interact with model outputs, and the degree of success that a model achieves in that regard is a crucial component of any evaluation.

Vega Health applauds HHS and ASTP/ONC for their time and attention towards incentivizing and promoting AI capable of safely and securely improving patient outcomes, reducing the cost of care, and enhancing clinical workflows. To best support

the refinement of clinical AI evaluation science and adoption, HHS should draw lessons from some of the most successful private sector models currently in operation.

We must ensure that widespread digital maturity and AI adoption across the healthcare sector does not exacerbate a digital divide between the best-resourced academic medical centers, and the community health centers that serve a high percentage of the U.S. population. The leading academic medical centers and health systems in the United States are centers of innovation and early adopters capable of developing and piloting transformative technologies. However, these systems are able to innovate and experiment due to a degree of financial security that most health systems lack. Financial headwinds, poor payer mix, and shrinking margins have left many healthcare providers – particularly rural and Critical Access Hospitals – without the luxury of launching innovation hubs, incubators, and competitive pilot programs.

Vega Health co-founder and CEO Mark Sendak has been at the forefront of addressing this challenge through the [Health AI Partnership](#), a multi-stakeholder collaborative that draws on educational materials, guidance, policies, and evaluations from leading U.S. health systems to eliminate the digital divide and build AI capacity in the long tail of healthcare.

Vega Health operates under these same principles to both accelerate innovation at cutting-edge health systems, and make those same AI technologies available affordably, safely, and securely for health systems and hospitals across the United States. HHS should invest in hub-and-spoke programs that scale localized innovation and evaluations to elevate the quality and reduce the cost of care nationally. Similar to the design of the Rural Health Transformation Program, HHS should design grant funding or competitive programs that incentivize AI evaluation and innovation to incorporate rural, critical access, and community health providers. Taxpayer-funded innovation should broadly benefit the American people, not simply reimburse innovation that is already funded. For example, HHS could require that models evaluated on specific patient populations or designed in leading health systems are easily scalable or have distribution strategies that will make them broadly accessible and accelerate adoption nationally.

Finally, the ROI used to evaluate clinical AI worth scaling must be tied to clinical outcomes. The Vega Health Marketplace contains models that have been evaluated in clinical settings across several use cases, including:

- Behavioral emergency and workplace violence
- Cardiac deterioration

- ED triage to inpatient beds
- ED triage to ICU beds
- ICU patient hand-off
- Low-risk pulmonary embolism for hospital discharge
- Mortality risk during encounter
- Post-surgical complication risk prediction
- Sepsis
- Unanticipated deterioration to ICU
- 48-hour mortality

At Duke Health and other health systems, these solutions have driven real-world improvements in clinician efficiency, quality of patient care, timeliness of patient care, and improved management of health system bed capacity.

Effective real-world monitoring requires moving beyond narrow technical metrics to assess AI performance across multiple, interconnected dimensions. Based on our experience, Vega Health is committed to a four-dimensional monitoring framework that captures whether AI solutions deliver sustained value in practice, and we urge ASTP/ONC to continue removing regulatory barriers across the following components of evaluation and monitoring.

Technical Accuracy and Model Fidelity. Technical performance remains the foundation of AI monitoring, but real-world technical assessment differs substantially from pre-market validation. In dynamic clinical environments, Vega Health monitors for model drift, including changes in model inputs, outputs, or the relationship between them that signal potential degradation. Monitoring data quality of model inputs is particularly important. Vega Health utilizes the Machine Learning-Data Quality Assurance (ML-DQA) [framework](#) developed at DIHI to assess completeness, conformance, and plausibility of all model inputs. Finally, Vega Health conducts retrospective and prospective evaluation of AI solutions in a secure development environment before going live in a clinical setting, ensuring model performance is satisfactory and preventing safety incidents.

User Adoption and Clinical Integration

Technical accuracy means very little if clinicians or end-users do not use AI tools appropriately, or if the tools disrupt established clinical workflows. Vega Health measures override rates, time-to-action on AI-generated alerts, and patterns of engagement that reveal whether tools provide value or generate alert fatigue to reveal critical insights that technical monitoring may miss. For example, high override rates

may indicate low positive predictive value prompting alarm fatigue, or they may reveal workflow misalignment and/or insufficient user training. The infrastructure to support adoption monitoring includes integration with clinical workflow systems to capture interaction data, mechanisms for collecting structured user feedback, and analytics capabilities that can identify patterns across different clinical contexts and user groups.

Clinical and Operational Outcomes

The third monitoring dimension assesses whether AI systems achieve their intended clinical and operational objectives. Technical accuracy and user adoption are necessary, but insufficient. The central question is whether an AI solution improves patient outcomes or organizational performance. This requires connecting AI integration to downstream outcomes. Measuring outcome impact presents methodological challenges, and the metrics tracked will vary significantly based on the nature of the AI solution or AI-enabled device. By integrating AI performance data with electronic health record (EHR) data and operational systems, Vega Health tracks patient outcomes, operational efficiency metrics, and financial performance in relation to AI implementation and usage patterns.

Return on Investment and Value Realization

The business case for AI monitoring itself becomes clear: without systematic measurement, healthcare delivery organizations cannot distinguish successful investments from expensive failures. On the benefit side, we seek to track both tangible returns, and harder-to-quantify improvements such as clinician satisfaction, faster decision-making, and reduced cognitive burden. ROI monitoring must account for the full lifecycle of AI implementation. Initial investments in integration and training may exceed short-term benefits, making it critical to track value realization over appropriate time horizons. We have observed that AI solutions often deliver value differently than anticipated – a tool designed to improve diagnostic accuracy may deliver value through workflow efficiency gains, or vice versa. Comprehensive monitoring captures these unexpected benefits while identifying solutions that fail to deliver value.

Comprehensive Monitoring in Practice

The four dimensions outlined above operate interdependently. Technical drift may manifest first as changing adoption patterns or declining outcomes. Poor workflow integration may appear as low adoption rates or negative user feedback before impacting clinical outcomes. Effective monitoring requires tracking all dimensions, with specific metrics tuned to the specific use case, and governance structures that can

synthesize insights across them to inform decisions about continuing, adapting, or sunsetting an AI solution.

The monitoring framework described above is not aspirational. It is derived from practical implementation experience and refined through implementation across diverse healthcare settings. The question is not whether comprehensive monitoring is necessary, but how health systems will build or access the infrastructure and expertise to do it effectively. As ASTP/ONC forges ahead with a new approach to regulating health IT, including novel AI applications in clinical and administrative settings, it is essential that the Assistant Secretary understand current capabilities and best practices. Deregulatory efforts do not make patients less safe or health systems less prepared; rather, they allow innovative AI solutions to take cost out of care, improve broken processes, and put patient outcomes first.

VI. Promoting Interoperability (Question 8)

In the recent HTI-5 NPRM, ASTP/ONC rightly acknowledges continued reports of leading EHR vendors leveraging scale and market dominance in the database and data management layer to gain a competitive advantage in the application layer. These actions amount to a weaponization of health IT certification standards and processes to stifle innovation and competition, to the detriment of the broader healthcare ecosystem and patients. EHR vendors have made it technically difficult and costly for health systems themselves to access and use health data generated by their providers and patients. These obstacles are amplified for health systems that want to utilize additional vendors bringing innovative technologies like AI. Nearly two decades ago, lawmakers required and funded the adoption of EHRs across healthcare under the expectation that an interoperable database layer would accelerate the pace of innovation, improve the quality of care for patients, and make healthcare data usable. HTI-5 has the potential to bring that promise closer to reality by ending these practices.

Today, EHRs operate not only as health record databases but as healthcare application developers. In reality, this has created a fundamentally misaligned incentive where the technology vendors controlling health system data are granted broad leeway to impose technical barriers, access fees, and administrative requirements on their competitors. These practices stifle technological innovation in healthcare, drive up the cost of new technologies for patients and providers, and contribute to the [inflationary effect](#) of technology in healthcare.

Vega Health is encouraged by HHS's [announced crackdown](#) on information blocking. The loosely defined nature of information blocking exceptions has allowed well-financed actors to simply reject interoperability requests and force lengthy, costly litigation.

Information blocking was intended to make interoperability the default state, with actors required to prove a request fell under an exception. Instead, dominant incumbents have abused exceptions to the rule to require innovative patient- and provider-facing technologies to prove that a request should be fulfilled.

In the absence of HHS enforcement actions to establish precedent and better define exceptions, certified health IT vendors have shifted the burden and cost of proving information blocking to innovators. Vega Health strongly supports efforts to eliminate or tighten frequently abused exception-based information blocking loopholes.

Over the past decade, it has also become apparent that FHIR APIs are a crucial step towards healthcare interoperability and standardized data exchange. FHIR-based data exchange provides a roadmap for the kind of real-time data exchange necessary to build meaningful AI and machine learning applications that leverage structured and unstructured data in healthcare. As ASTP/ONC has frequently acknowledged, the future of healthcare interoperability and data usability will be built on FHIR standards.

As ASTP/ONC continues to develop both the FHIR standard for data exchange and the US Core for Data Interoperability (USCDI) definitions of crucial data elements, Vega Health encourages policymakers to prioritize standardizing elements essential to the implementation of AI/ML solutions in health systems. From experience implementing AI solutions at multiple health systems, Vega Health has recognized that Medication Administration Records are crucial inputs for commonly used clinical AI solutions, such as inpatient deterioration or emergency department likelihood to admit predictive models. Today, MAR data must be accessed from external databases, through custom API infrastructure, or via HL7 data feeds – each approach complicating model implementation and usability. Vega Health is encouraged by the inclusion of MAR data in the draft USCDI v7 and encourages ASTP/ONC to continue expanding the set of standard data elements as a key measure to accelerating the implementation and use of clinical AI solutions.

Additionally, for FHIR APIs to sustain and scale health AI innovation, a few key roadblocks must be resolved. In the HTI-5 NPRM, ASTP/ONC proposed a key regulatory shift in the rulemaking authority derived from the 21st Century Cures Act and sets the foundation for future API-based certification criteria. Vega Health applauds this shift and encourages ASTP/ONC to use these criteria to set baseline technical standards for APIs that can address some of the information blocking practices described above. Vega Health recommends that ASTP/ONC take the following suggestions under consideration when developing API-based certification criteria:

Vega Health urges ASTP/ONC to require reasonable, market-based pricing for data queries. Often, certified health IT vendors restrict the use of FHIR APIs by placing prohibitive user fees or per-query fees on developers. These prices far exceed any API development or maintenance costs that the certified health IT vendor may incur and instead are instituted to suppress and restrict innovative technologies that a health system may seek to deploy. The changes to the manner exception in the HTI-5 NPRM acknowledge this challenge broadly, but ASTP/ONC should preemptively eliminate the same practices as they apply to API query fees specifically.

Vega Health urges ASTP/ONC to place limits on the ability of certified health IT vendors to cap the number of queries or flow of data through FHIR APIs. It should not be up to data-layer technologies that health systems are required to utilize by federal law to set limits on the ways in which health systems and their chosen vendors can utilize or access system data. Complicating the picture, many certified health IT vendors develop their own applications that are in direct competition with outside vendors; the ability to place restrictions on competitors that do not extend to their own enterprise solutions gives certified health IT an anticompetitive advantage and creates misaligned incentives, limiting innovation and competition. While some limits may be technically necessary, these limits should be set via documented API specifications, health system or end user throughput limitations and cost containment, or contractual agreement, not imposed inconsistently by certified health IT to gain a competitive advantage.

Vega Health urges ASTP/ONC to establish and reinforce “right to connect” and “right to build” permissions for health systems using certified health IT. As part of API-based Health IT Certification Program criteria, ASTP/ONC should ensure that health systems have the ability to access and utilize FHIR APIs for AI implementations and development projects. Health systems have frequently encountered obstacles to internal AI development or real-time data analysis when specific data elements are unavailable through required certified health IT APIs.

Beyond common sets of data elements required by law or embedded into standards like the US Core for Data Interoperability, certified health IT vendors place restrictions or add multiple burdensome hurdles to prevent health systems and their chosen vendors from building custom APIs that fully utilize health system data. These barriers include contractual language that limits the ability of AI solution vendors to employ or sub-contract domain experts with the relevant expertise to effectively integrate AI solutions into legacy health IT systems. Legacy health IT incumbents are limiting innovation and competition by preventing new entrants from being able to bring in the relevant expertise to expand AI use in healthcare. This is antithetical to innovation,

anticompetitive, and an artificial inhibitor of AI innovation and ROI. A universal “right to build” permission would allow health systems and contracted vendors to take full advantage of FHIR APIs to unlock lower costs and improved patient outcomes through digital development.

Vega Health urges ASTP/ONC to set standard FHIR API configurations that are publicly available to innovators and entrepreneurs. By shifting the Health IT Certification Program to API-based criteria, ASTP/ONC has the opportunity to eliminate the lack of standardization and technical hurdles that slow and complicate implementation projects, adding unnecessary cost to healthcare. Uniform, universal API documentation will accelerate innovation, make data and analytics insights more widely accessible for lower-resourced health systems, and create more resilient healthcare data systems for a more connected future state of healthcare.

VII. Clinical AI Research (Question 10)

In order to accelerate the adoption of clinical AI, HHS should prioritize simplifying the process for and reducing the cost of testing new AI models quickly across diverse patient populations and clinical settings. The FDA’s process for reviewing new drugs and medical devices is costly and time-intensive, largely due to challenges with designing clinical trials, identifying trial participants, and conducting trials either on-premise or from a decentralized hub. However, testing AI solutions – systems built on algorithm-based models that can be quickly tested on large datasets and tuned to achieve desired results – can and should look fundamentally different.

Safety is a top concern for clinical AI applications. Whenever a new technology influences clinician decisions and patient outcomes, it is imperative that performance is predictable, controllable, and reliable. In order to ensure clinical AI safety and performance, HHS should invest in research that improves both the timeliness and representativeness of AI evaluation and testing.

Another significant consideration for the testing of new AI solutions is patient privacy. Federal and state law protects the privacy of medical information, and patients have a reasonable expectation that their data will not be exposed to researchers or used to train or evaluate AI models without their consent.

Vega Health’s unique approach to AI development and implementation places models directly into health system networks, limiting data exfiltration and making patient records invisible to model developers and researchers. By standardizing data element and variable definitions, Vega Health accelerates AI model pilots in clinical settings. Any health system that has implemented the Vega Health Platform can quickly connect new

AI models, regardless of where the model was developed or what variable definition the model uses. Once a model is implemented onto the Vega Health Platform, it could be evaluated through shadow pilots on real-world patient cohorts in a wide range of clinical settings – all without ever exposing patient data to researchers, developers, or anyone outside of the health system where the patient already receives care. This process would also give health systems more autonomy and insight into how AI models are tested – by giving health systems the ability to determine which models to evaluate, consent to evaluation on their data, and provide visibility into any testing, HHS can rapidly accelerate the pace of clinical AI research.

The feasibility and effectiveness of these silent trials has been [established](#) by researchers at leading institutions, including Vega Health's co-founder and CEO. HHS should leverage these learnings and invest in increasing access to shadow pilots that protect patient privacy, ensure model safety and performance before go-live, and accelerate the pace of adoption of AI models in clinical settings across the country.

VIII. Conclusion

Vega Health appreciates ASTP/ONC's commitment to accelerating the use of safe, effective clinical AI solutions in healthcare. A whole-of-HHS approach will be necessary to effectively reduce barriers to adoption and incentivize innovation that improves patient outcomes and reduces the cost of delivering quality healthcare. If Vega Health can be a resource, please do not hesitate to reach out to Vega Health co-founder and CEO Mark Sendak (Mark.Sendak@vegahealth.com).

Sincerely,



Mark Sendak, MD, MPP

Co-founder and CEO, Vega Health